

v14 Research-Ready Upgrade: Memory, Readiness and Evidence-Grounded Copilot

Change log and implementation summary for supervisor/dissertation evidence

1. What was done before

- v13.3 already had modern dashboard navigation, local/URL/Google Drive/PostgreSQL-ready data loading, automatic modelling after upload, manual model training, visual analytics, business suggestions, Copilot chat, audit logging and export.
- The Copilot already answered from the active dataset and showed dataset source, columns, evidence, limitations and safety warning.
- Customer churn remained the main proof-of-concept case study, with other datasets used for robustness testing.

2. What changed now in v14

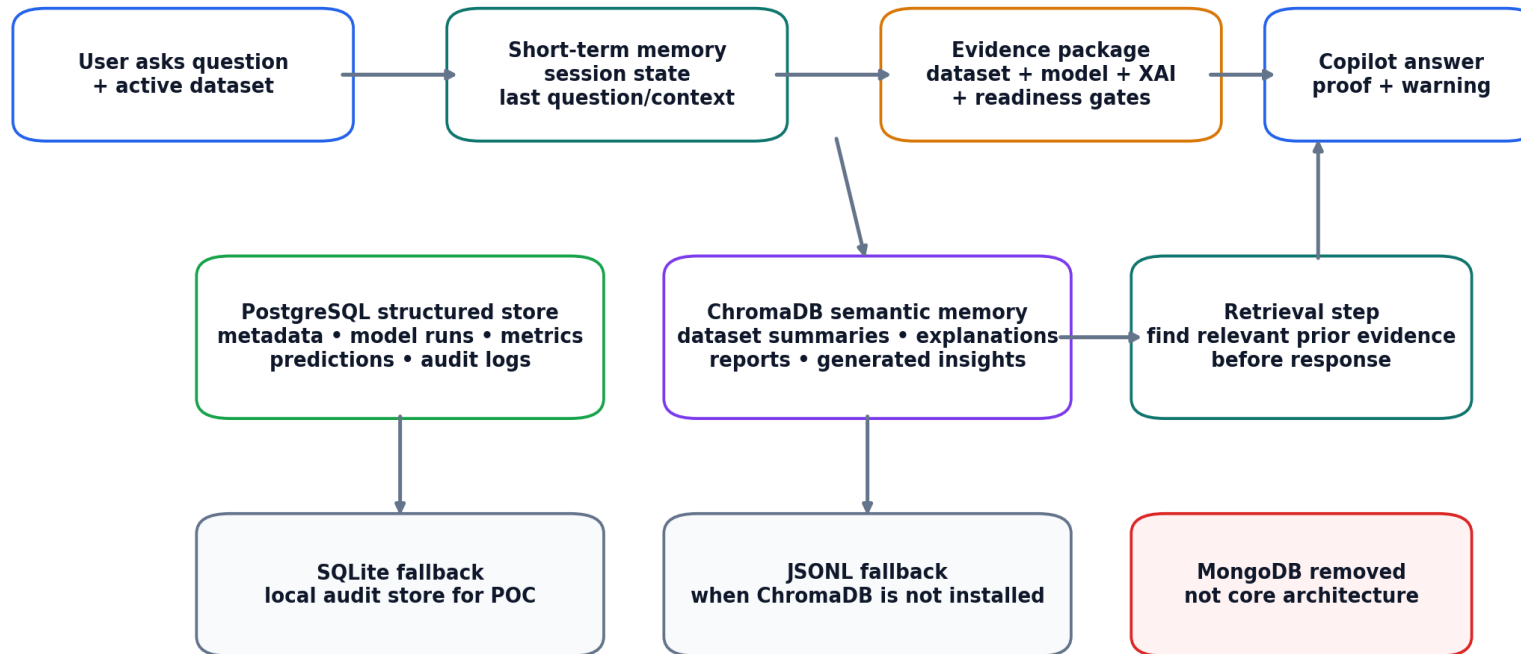
Area	v14 change
Research contribution	Added explicit contribution framework: readiness gates, evidence grounding, explainability-to-action, memory and evaluation.
Readiness gates	Defined measurable PASS/WARNING/FAIL thresholds for missingness, duplicates, row count, target readiness, class balance, privacy and evidence availability.
Memory	Added short-term session memory and ChromaDB semantic long-term memory with JSONL fallback.
Storage	Removed MongoDB from the core architecture. PostgreSQL remains for structured data; ChromaDB is used for semantic memory.
Copilot answer	Added answer-grounding status and evidence package logging.
Evaluation	Added model, explanation, grounding, trustworthiness, usability, decision-support and scalability metrics.
Export	Added research framework and semantic memory event export.

3. Memory architecture

The architecture now separates session context from persistent memory. Short-term memory supports follow-up questions in the active session. PostgreSQL is used for structured evidence and audit design. ChromaDB is used as long-term semantic memory for retrieving dataset summaries, explanations, audit records and generated insights. MongoDB is removed from the core design.

Figure 5. v14 Copilot Memory Architecture

Short-term session memory supports follow-up questions; PostgreSQL stores structured evidence; ChromaDB stores semantic long-term memory for retrieval-based grounding.



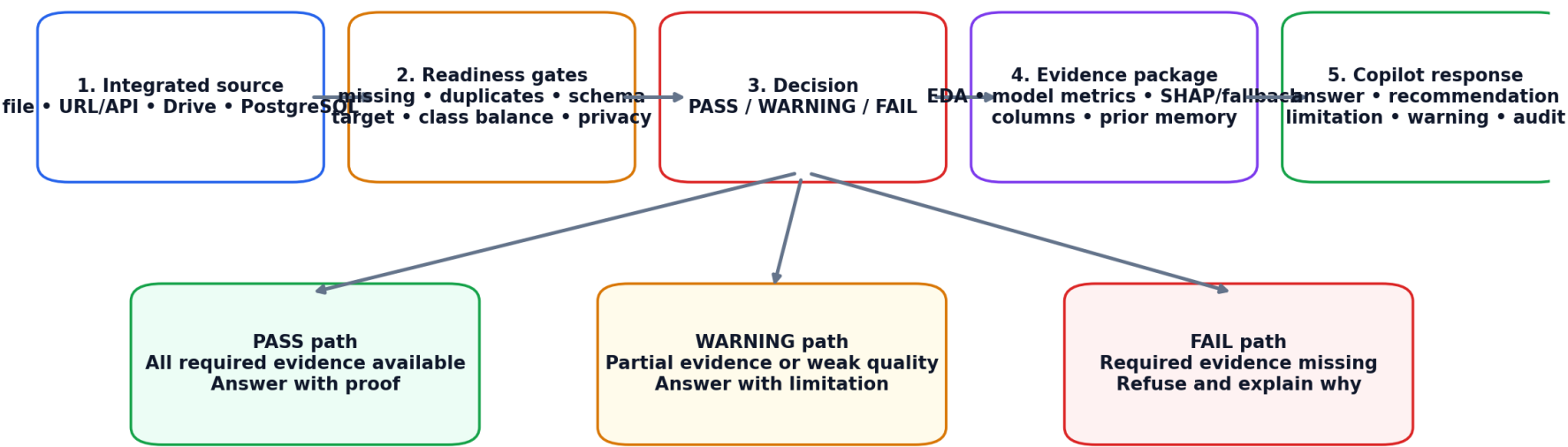
Memory purpose: maintain context, retrieve prior evidence, and make Copilot answers auditable and reproducible.

4. Readiness and answer-generation flow

The Copilot answers only after the system checks data readiness and retrieves a traceable evidence package. If required evidence is missing, the system explains why the answer cannot be produced reliably.

Figure 6. v14 Readiness and Evidence-Grounded Answer Flow

The Copilot answers only after data, target/model artefacts and question evidence pass measurable readiness and grounding checks.



Example FAIL: if the active Telco dataset does not contain balance, a bank-marketing question about balance must not be answered from churn data.

5. Files added or changed

File	Purpose
src/readiness_gates.py	Measurable PASS/WARNING/FAIL criteria.
src/research_contribution.py	Research contribution table and summary.

src/evidence_package.py	Traceable evidence package generation.
src/memory_manager.py	Short-term memory and memory architecture tables.
src/vector_memory.py	ChromaDB semantic memory with JSONL fallback.
src/evaluation_framework.py	Evaluation dimensions and metrics.
src/validation.py	Updated thresholds and readiness gate logic.
src/copilot.py	Added answer grounding status and aligned readiness wording.
app.py	Research & Memory page, ChromaDB memory UI, memory logging, MongoDB removed.

6. Testing result

Compile check passed: `python -m py_compile app.py src/*.py`. Unit tests passed: 25 passed.